



Multi-Vendor E-Commerce Marketplace

A unified web platform that connects independent vendors with customers, streamlines product management, and centralizes order processing.

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Problem & Motivation



Limited Reach for Small Sellers

Independent vendors struggle to find customers and scale without a central marketplace.



Fragmented Customer Experience

Customers must visit multiple sites to compare products, prices, and reviews — slowing decisions.



Inefficient Management

Vendors lack simple, centralized tools to manage stock, orders, and product updates.

A single platform reduces friction: centralized discovery, unified product data, and shared trust signals like ratings and reviews.



Abstract

The proposed Multi-Vendor E-Commerce Marketplace is a responsive web application that enables multiple sellers to list and manage products while providing customers with unified search, comparison, purchasing, and tracking workflows. Core features include vendor registration and product management, customer accounts and carts, order lifecycle tracking, and an administrator console to oversee users, vendors, products, and transactions.

By aggregating sellers and product catalogs, the system aims to increase vendor visibility, simplify customer decision-making, and provide scalable operations suitable for academic evaluation and real-world deployment.

Proposed Solution — Role-Based Features

Architectural Overview

A single codebase exposes RESTful APIs with role-based authorization. Roles: Vendor, Customer, Admin. Each role has tailored UI views and specific CRUD privileges to preserve data integrity and traceability.

- ❑ Use JWT for session authorization and bcrypt for secure password storage.

1

Vendor

Register, add & edit products, manage stock, view orders and fulfillment status.

2

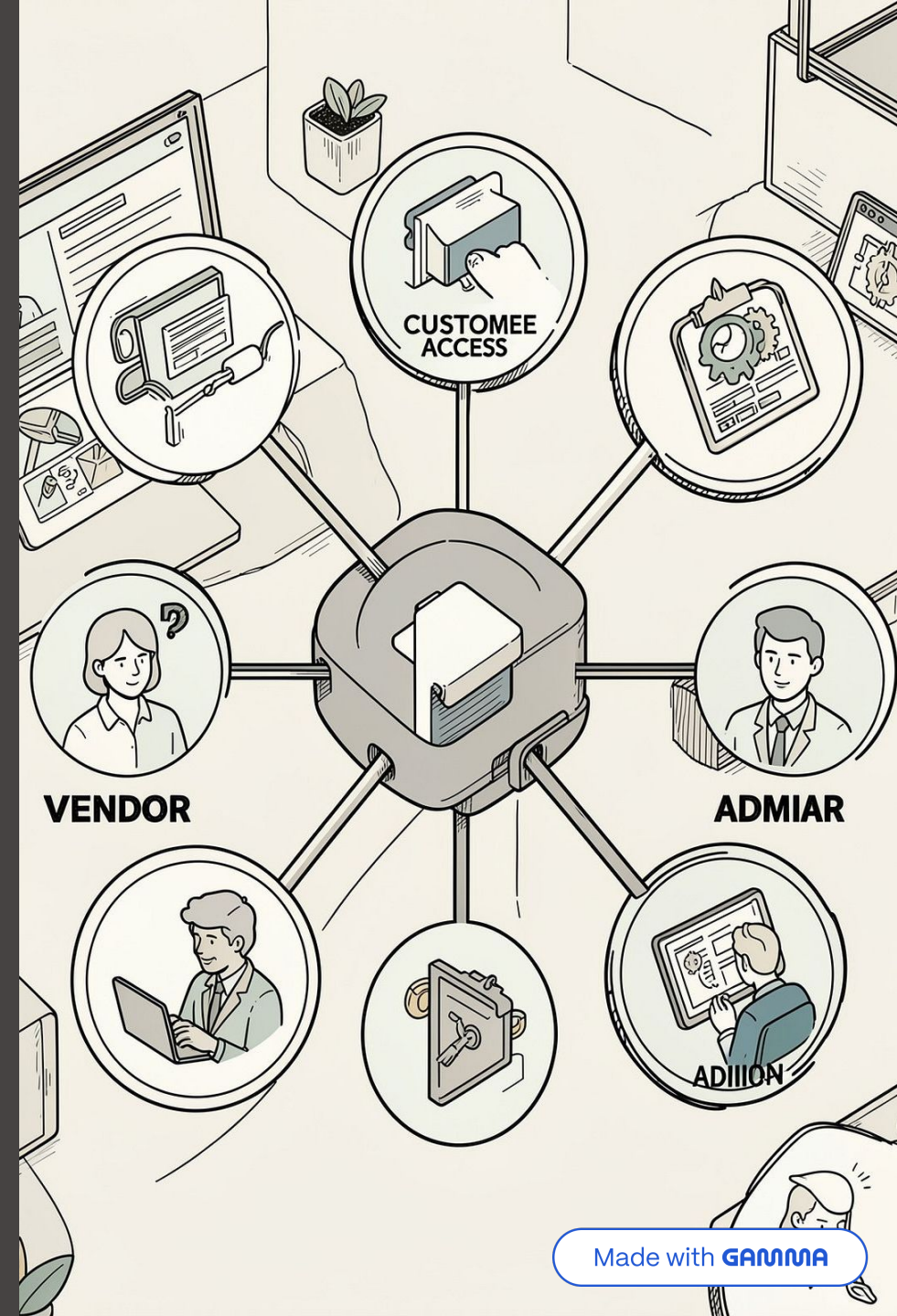
Customer

Browse & search catalogs, compare items, cart & checkout, track orders, rate products.

3

Admin

Manage users, onboard vendors, moderate products & reviews, monitor orders and reports.



Technology Stack



Frontend

React.js • Bootstrap •
CSS • Axios • React
Router — responsive,
componentized UI



Backend

Node.js • Express.js —
REST APIs,
authentication, and
business logic



Database

MongoDB with
MongoDB Compass
for schema
exploration and
flexible document
models



Authentication & Security

JWT tokens for
stateless auth,
bcrypt.js for password
hashing, role checks
on every endpoint



Development Tools

VS Code • Git •
GitHub — version
control, issue tracking,
and CI/CD readiness



Expected Outcome & Academic Evaluation Criteria

For Customers

Unified product discovery, seamless checkout flow, order tracking, and aggregated reviews facilitate faster, more informed purchases.

For Vendors

Simple onboarding, SKU and stock management, order visibility, and consolidated sales reports increase reach and operational efficiency.

For Admins & Instructors

Centralized control over users, vendors, and content; instrumentation for auditing, grading, and demonstration of DBMS concepts like schema design, indexing, and transaction handling.

Overall objective: deliver a robust, well-documented DBMS project that demonstrates practical application of web technologies, secure authentication, role-based access, and scalable data models.

Next steps: implement full API test suite, add payment gateway integration, and prepare demo data for evaluation.